

Sabbir Ahmed

✉ Email 📞 +880 1703184419 in LinkedIn 🌐 Website

Education

Bangladesh University of Engineering and Technology (BUET)

2020–2026

B.Sc. in Electrical and Electronic Engineering (EEE)

- CGPA: 3.57/4.00 (3.76 in last 2 terms)
- Class Rank: 102 out of 204 (Top 50%)
- Major: Communication and Signal Processing
- Thesis Title: A Novel Privacy-Preserving Algorithm for Membership Inference Test for Face Recognition [link](#) 

Research Interest

- Machine Learning in Sensing and Control
- AI Hardware
- Internet of Things (IoT)
- Embedded Systems

Research Experience

Bangladesh University of Engineering and Technology (BUET)

2026

Undergraduate Thesis

- Conducted research on privacy-preserving membership inference testing for face-recognition models, focusing on whether sensitive samples were used to train a frozen base model.
- Worked with facial-image preprocessing, ResNet100 activation-derived data, differential privacy and EfficientNet-B0-based classification for privacy-aware model auditing.

Publications

- **Machine Learning and IoT Based Automated Remote Health Monitoring and Alert System**, Md. Nagib Mahfuz, **Sabbir Ahmed**, Md. Shahriar Hasan, Md. Hamidur Rahman. Presented at the *2025 IEEE 4th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*; published in IEEE Xplore, 2026. [DOI]
- **Beyond Accuracy: An Empirical Risk Assessment of Shortcut and Segmentation Dependence in Chest X-Ray Classification**, **Sabbir Ahmed**, Kazi Ashiqur Rahman, Md. Nagib Mahfuz, Maruf Ahmed. Presented at the *2026 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Dhaka, Bangladesh, September 2026; publication in IEEE Xplore pending. [PDF]

Work Experience

Huawei Technologies (Bangladesh) Ltd.

2026–Present

Solution Engineer, South Asia Digital Power Business

- Working in solution engineering within Huawei's South Asia Digital Power Business, supporting technical solution activities in the digital-power domain.

Fiber @ Home Ltd.

2025

Industrial Attachment

- Gained practical exposure to optical-fiber communication, NTTN/IIG operations, DWDM systems, IP/MPLS routing, network planning, fault localization and redundancy planning.
- Visited Tier III data-center facilities and learned about precision cooling, infrastructure maintenance, power backup and service reliability.

Team Interplanetar – BUET Mars Rover Team

2022–2026

Team Leader, Video Editing Team; Documentation Sub-Team Leadership

- Led technical documentation and visual-content preparation for robotics and rover-related team activities.

Relevant Coursework

- Artificial Intelligence and Machine Learning
- Microprocessors and Embedded Systems
- Robotics and Automation
- Communication Systems I & II
- Digital Signal Processing I
- Wireless Communications
- Random Signals and Processes
- Control Systems I
- Continuous Signals and Linear Systems
- Digital Electronics
- Introduction to Medical Imaging
- Optical Communications

Academic Projects

Personal Assistive Robot for a Person with Speech and Hearing Impairment [link](#)

- Developed an autonomous assistive robot using Raspberry Pi Zero 2W and ESP32, with Bengali sign-language-to-speech, speech-to-text, computer vision and person-following functionality.
- **Tools Used:** Python, TensorFlow Lite, OpenCV, Raspberry Pi, ESP32

IEEE Signal Processing Cup 2026: Real-Time Audio Visual Zooming for Smartphones [link](#)

- Developed an audio-enhancement pipeline using MVDR beamforming, IPD masking, WPE dereverberation, Wiener filtering and deep-learning post-processing; achieved 7.85 dB OSINR improvement.
- **Tools Used:** MATLAB, Python, Digital Signal Processing, PyTorch

Smart Traffic Governance with Strategic AI Deployment [link](#)

- Implemented a computer-vision-based traffic governance system for illegal-parking detection, automated ticket generation and emergency-vehicle path clearing; first-prize project at Impact Dhaka Hackathon 2026.
- **Tools Used:** Computer Vision, Edge AI, Python, Geospatial Mapping

IoT Based Remote Patient Health Monitoring System [link](#)

- Integrated ESP32, biomedical sensors, cloud monitoring and alerts with an ECG classifier based on a 1D CNN for continuous remote patient monitoring.
- **Tools Used:** ESP32, IoT, TensorFlow, Python, Sensors

Machine Learning Based Automated Bandwidth Analysis and Estimation [link](#)

- Co-developed a time-series-based system to estimate bandwidth requirements for a location at different times of the day using multiple forecasting models.
- **Tools Used:** Python, TensorFlow, Time Series Analysis, Data Analysis

Automated Medical Waste Segregation Using Computer Vision [link](#)

- Built an automated segregation system using YOLO-based object detection and embedded control to identify and route medical waste into appropriate bins.
- **Tools Used:** YOLOv8/YOLO11, OpenCV, Python, C++, Embedded Systems

Digital Electronics Circuit Based Tic-Tac-Toe [link](#)

- Designed and implemented a two-player Tic-Tac-Toe system using digital logic, custom PCB hardware, turn-control logic, win detection and LED indicators.
- **Tools Used:** Digital Logic Design, PCB Design, Proteus

Technical Skills

Programming Languages: Python, C, C++, MATLAB

AI/ML and Data Analysis: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, NumPy, Pandas

Embedded Systems and IoT: ESP32, STM32, Arduino, Raspberry Pi, sensor interfacing, cloud-based monitoring

Engineering and Simulation: MATLAB/Simulink, PSpice, LTspice, Proteus, AutoCAD, Altium

Telecom and Networking: Optical fiber communication, DWDM, IP/MPLS routing, NTTN/IIG operations

Achievements & Awards

- **Impact Dhaka Hackathon 2026 – 1st Prize**
- **HULT Prize at BUET 2022 – 1st Runners Up**
- **Dean's List**, Bangladesh University of Engineering and Technology